

Millsorpt – Mill Scale to Metal-Free: Converting Steel Mill Waste into Heavy-Metal Adsorbents

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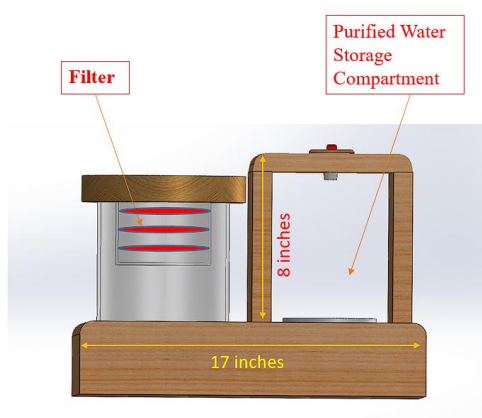
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Abstract

Industrial steel processing generates large amounts of ferrous mill scale, a potential source of iron oxide, which is usually disposed of. In this capstone project, we converted mill scale into a chemically-treated iron oxide-based adsorbent material and created a low-cost filtration system using the mill scale-based adsorbent for removing Cu, Ni, and Cr from industrial process wastewater effluents. Characterization using x-ray diffraction (XRD) confirmed that the iron-oxide phases were ideal for adsorption. The processed adsorbent, when fitted into a laboratory-scale filter column, was found to be efficient in removing metals from binary and mixed solutions (i.e., up to ~95.8%) when analyzed by atomic absorption spectroscopy (AAS) and ultraviolet-visible spectroscopy (UV-vis). **The project is expected to achieve three goals: valorization of industrial waste, cleaning of industrial toxic wastewater, and metal recovery with economic value at low costs.**

Methodology

Mill scale was taken from the local steel industry's waste, ground to a powder, and the iron oxide phases were analyzed by X-ray diffraction (XRD). The resulting powder was treated with H_2SO_4 , neutralized with NaOH , washed and dried several times to produce the hematite-rich adsorbent. Finally, a simple bench-scale filtration unit was constructed with layers of pebbles, sand, charcoal and the processed adsorbent as the active layer. Solutions of heavy metals (Cu, Cr and mixed Cu-Cr-Ni species) were filtered through the set-up to test the efficacy of the adsorbent. The quantities of removed Cu, Cr, and Ni were quantified by AAS (with UV-vis spectroscopy mainly being used to confirm the Cu adsorption efficiency).



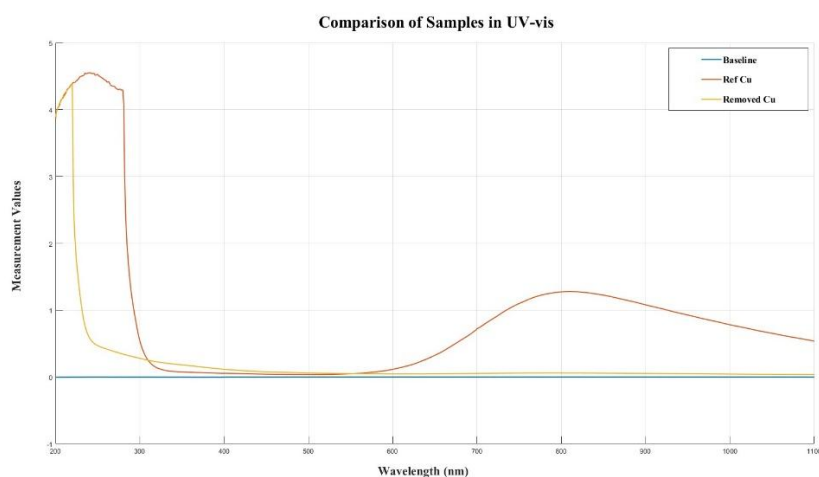
Our Product

Result

From AAS test:

Sample type	Heavy metal(s)	Removal after treatment (%)
Single-metal solution	Cu	72.78
Single-metal solution	Cr	76.62
Mixed-metal solution	Cu (mixed)	89.00
Mixed-metal solution	Cr (mixed)	90.00
Mixed-metal solution	Ni (mixed)	95.76

- UV-vis spectra of Cu solutions showed a marked decrease in absorbance peaks after filtration, confirming effective adsorption of Cu^{2+} by the mill-scale-derived material.
- The prototype filter operated successfully in multiple trial-and-error runs, demonstrating stable flow and reproducible removal performance.



Conclusion

Mill scale is a byproduct of steel processing. It can be used to generate iron-oxide adsorbents that can effectively remove heavy metals (Cu, Cr, Ni) from contaminated water. Using an adsorbent obtained from mill scale, the removal of contaminants was performed using a lab-scale filtration unit. Removal efficiencies of ~95.8% were achieved using mixed-metal contaminants. The technology uses a waste product as an adsorbent to treat water contaminated with heavy metals and enables recovery of precious metals from the adsorbent. Steel industries and other national research institutes gave very positive feedback in respect of industrial relevance and market readiness. Future work will further investigate mechanical strength of adsorbent in the form of pellets or tablets. Other aspects include optimizing chemical modification for improved adsorption capacity, and scaling the filtration unit for continuous operation in the treatment of real industrial wastewaters to realize the potential of this sustainable mill scale-based water treatment technology.